

LECTURE 2 — SKILL OF THE WEEK: NOTE- TAKING

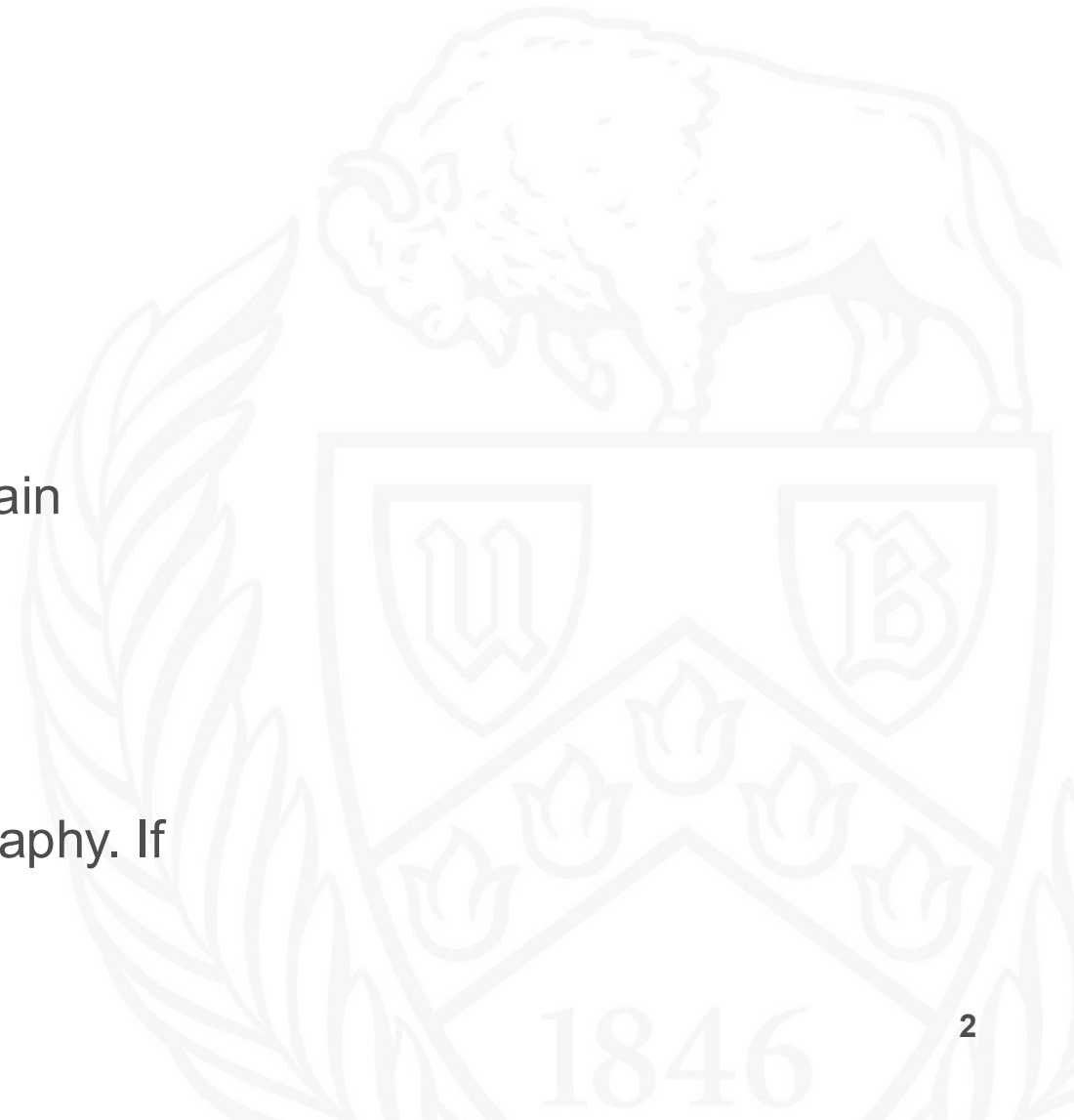
PHY199-RAP · Energy in the 21st Century

Dr. Salvatore Rappoccio



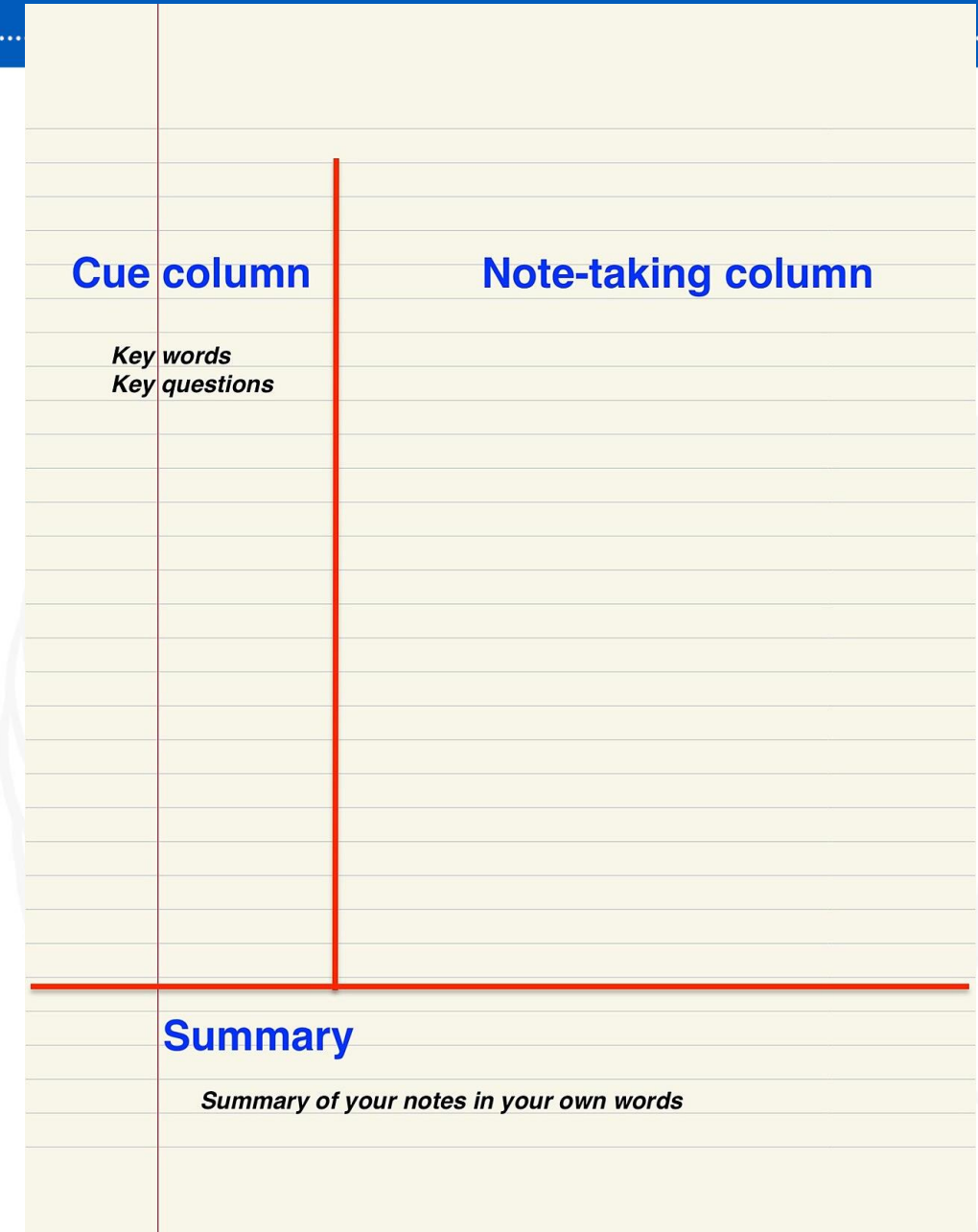
The skills to succeed in college

- This is a **UB Seminar** so we teach “how to college”
 - This week: **taking notes**
- Use your brain! **Process, don’t transcribe**
- Writing things down helps them be cemented in your brain
 - Some science behind this:
 - Mueller & Oppenheimer, 2014
 - Weak replication in Urry et al., 2021
 - The real lesson: notes are for **thinking**, not stenography. If you’re just copying, you’ve turned your **brain off**



The Cornell method

- **1950s, Cornell:** Walter Pauk designed a page layout that forces processing
 - “How to Study in College”
- Split the page: **narrow left column** (cues/questions), **wide right column** (notes in class), **strip along the bottom** (a 2-line summary you write *after*)
- Can cover the right side, and your left-column questions become a **self-quiz**
- It bakes in the two things that actually work (next slide): **retrieval** and **spacing**



What the evidence says actually works

- Two techniques that are demonstrated to work: **practice** and **spacing**
 - Be consistent, let your prefrontal cortex be engaged!
 - When you're panicking, you don't think, reflexes take over
 - Consistent regular practice cements things in your **rational mind**
- **Rereading** and **highlighting** *feel* productive but doesn't engage your mind
- In this class:
 - Take notes you can be **quizzed from**, then actually quiz yourself
 - We're mostly focused on vocab, history, a few concepts of science/engineering
- In other classes:
 - Keep practicing example problems

Practice
makes
permanent!

Derivations: Special Case

- In many classes, you'll perhaps be asked to follow along with derivations of mathematical formalisms (math, physics, chemistry, engineering)
- Can adapt the Cornell method to use the left column as a "narration" area so you can understand what's going on later

Force is
time rate
of change
of
momentum
(mass $\times$ velocity)

use chain
rule!

$$\vec{F} = \frac{d\vec{p}}{dt}$$

but $\vec{p} = m\vec{v}$

so $\frac{d\vec{p}}{dt} = \frac{dm}{dt}\vec{v} + m\frac{d\vec{v}}{dt}$

if $\frac{dm}{dt} = 0,$

$$\frac{d\vec{p}}{dt} = m\frac{d\vec{v}}{dt} = m\vec{a}$$

For systems that don't
change, $\vec{F} = m\vec{a}$

Example: Should you wash your rice?

- We'll do the Cornell method together
- Fill the **right column** first, but **don't just transcribe**
- The **left column** and the **summary** come during a pause or after class
- Let's watch this together:
 - @pbsterra [short on washing rice](#)
 - [The Conversation](#) from Evangeline Mantzioris, Program Director of Nutrition and Food Sciences University of South Australia
 - Yu et al.: [Review of the effects of different processing technologies on cooked and convenience rice quality](#)
 - Dessi et al.: [Plastics contamination of store-bought rice](#)

Review

Review of the effects of different processing technologies on cooked and convenience rice quality

L. Yu ^a  , M.S. Turner ^{a b} , M. Fitzgerald ^a , J.R. Stokes ^c , T. Witt ^a 

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<https://doi.org/10.1016/j.tifs.2016.11.009> ↗

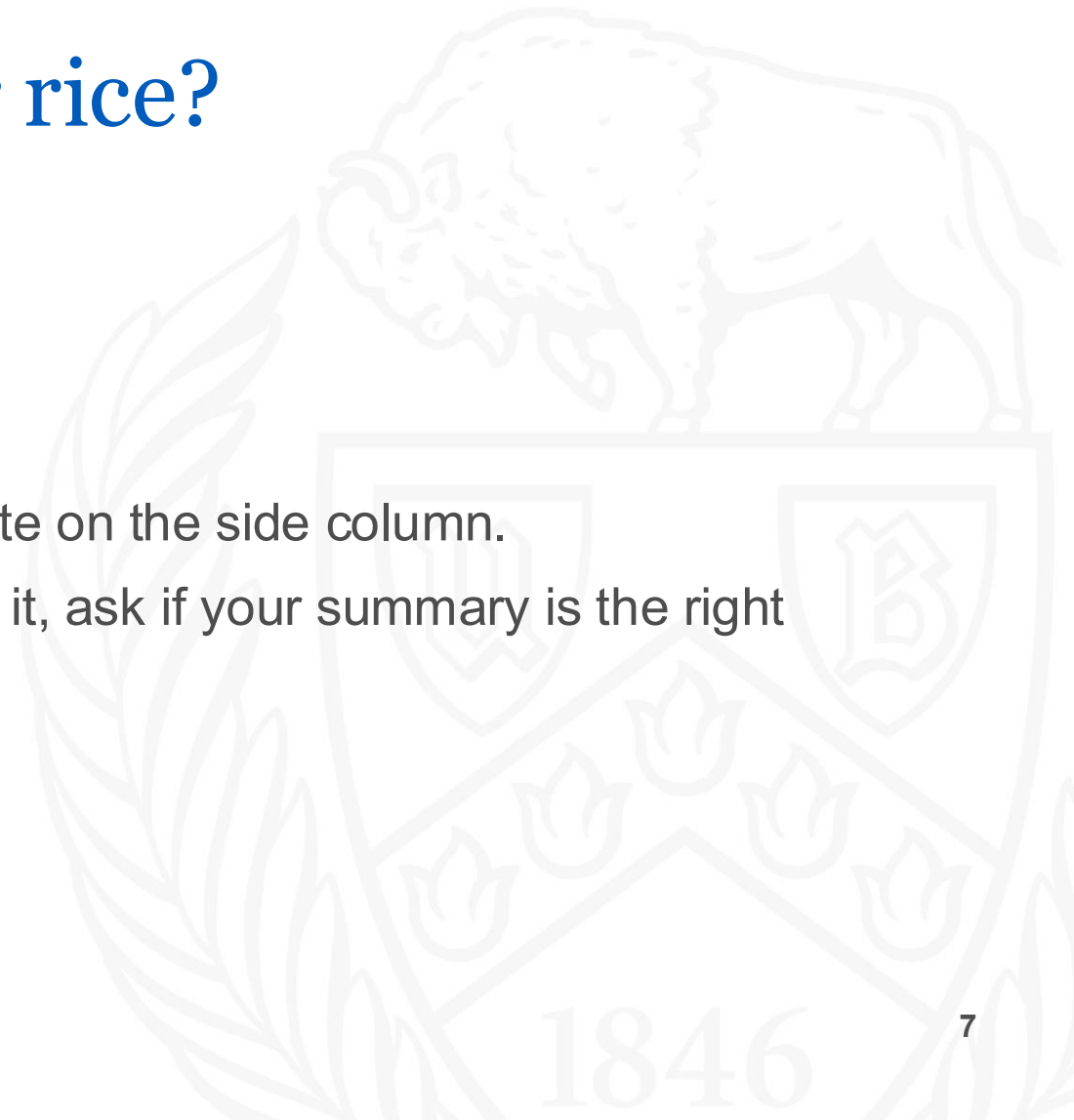
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Highlights

- Processing related reduction of convenience rice quality is mechanistically explained.
- Flavour deteriorates as volatile profiles are altered by thermal processes and storage.
- Cooked rice texture is dependant on the rate and extent of water and starch diffusion.
- Different mechanisms in high pressure processes may improve convenience rice quality.

Example: Should you wash your rice?

- Hints:
 - Use abbreviations
 - Draw arrows between concepts
 - If something confuses you, ask! Then jot down a note on the side column.
 - At the end, summarize, and if you want to really get it, ask if your summary is the right understanding



Example: Should you wash your rice?

Raw notes:

- Washing rice controversial on interwebs
- Affect stickiness? Studies say no.
- TYPE of rice matters, washing doesn't
- Health:
 - remove 20% of microplastics, also arsenic
 - but washes away copper, iron, zinc (nutrients!)
- Doesn't remove bacteria

Example: Should you wash your rice?

Cue Column:

Does washing affect stickiness?

Pros of washing rice

Cons of washing rice

Misconceptions about washing

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Summary:

Washing doesn't affect stickiness. Contaminants removed but also nutrients.

Note-taking — sources

- [Mueller & Oppenheimer \(2014\), *Psychological Science*](#) · [Urry et al. \(2021\) — replication](#)
- [Dunlosky et al. \(2013\), *Psych. Science in the Public Interest*](#) · [accessible version, *American Educator*](#)
- Cornell method: Pauk & Owens, *How to Study in College*; [Cornell Learning Strategies Center](#)
- Worked example (washing rice): [The Conversation — “Do you need to wash rice? Here’s the science”](#) · arsenic/nutrient tradeoff: [“Cooking rice in excess water reduces both arsenic and enriched vitamins...” \(2016\); Menon et al. 2020, *Sci. Total Environ.*](#)